

final-project-203a

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```
library(tidyverse)
library(janitor)
library(nnet)
library(randomForest)
library(skimr)
library(tidymodels)
library(GGally)

tidymodels::tidymodels_prefer()
set.seed(20251212)

# Read data
fetal_raw <- read_csv("fetal_health.csv")

# Clean column names: 'baseline value' -> baseline_value, etc.
fetal <- fetal_raw %>%
  clean_names() %>%
  mutate(
    fetal_health = factor(
      fetal_health,
      levels = c(1, 2, 3),
      labels = c("Normal", "Suspect", "Pathological")
    )
  )

glimpse(fetal)

## Rows: 2,126
## Columns: 22
## $ baseline_value          <dbl> 120, 132, 133, ~
## $ accelerations           <dbl> 0.000, 0.006, 0~
## $ fetal_movement          <dbl> 0.000, 0.000, 0~
## $ uterine_contractions    <dbl> 0.000, 0.006, 0~
## $ light_decelerations     <dbl> 0.000, 0.003, 0~
## $ severe_decelerations    <dbl> 0, 0, 0, 0, 0, ~
## $ prolonged_decelerations <dbl> 0.000, 0.000, 0~
## $ abnormal_short_term_variability <dbl> 73, 17, 16, 16,~
## $ mean_value_of_short_term_variability <dbl> 0.5, 2.1, 2.1, ~
## $ percentage_of_time_with_abnormal_long_term_variability <dbl> 43, 0, 0, 0, 0,~
## $ mean_value_of_long_term_variability <dbl> 2.4, 10.4, 13.4~
## $ histogram_width         <dbl> 64, 130, 130, 1~
## $ histogram_min           <dbl> 62, 68, 68, 53,~
## $ histogram_max           <dbl> 126, 198, 198, ~
```

```
## $ histogram_number_of_peaks      <dbl> 2, 6, 5, 11, 9, ~
## $ histogram_number_of_zeroes     <dbl> 0, 1, 1, 0, 0, ~
## $ histogram_mode                 <dbl> 120, 141, 141, ~
## $ histogram_mean                 <dbl> 137, 136, 135, ~
## $ histogram_median               <dbl> 121, 140, 138, ~
## $ histogram_variance             <dbl> 73, 12, 13, 13, ~
## $ histogram_tendency             <dbl> 1, 0, 0, 1, 1, ~
## $ fetal_health                   <fct> Suspect, Normal~
```

```
# missingness check
fetal %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "var", values_to = "n_missing") %>%
  arrange(desc(n_missing)) %>%
  print(n = 50)
```

```
## # A tibble: 22 x 2
##   var                n_missing
##   <chr>              <int>
## 1 baseline_value      0
## 2 accelerations      0
## 3 fetal_movement     0
## 4 uterine_contractions 0
## 5 light_decelerations 0
## 6 severe_decelerations 0
## 7 prolonged_decelerations 0
## 8 abnormal_short_term_variability 0
## 9 mean_value_of_short_term_variability 0
## 10 percentage_of_time_with_abnormal_long_term_variability 0
## 11 mean_value_of_long_term_variability 0
## 12 histogram_width    0
## 13 histogram_min      0
## 14 histogram_max      0
## 15 histogram_number_of_peaks 0
## 16 histogram_number_of_zeroes 0
## 17 histogram_mode     0
## 18 histogram_mean     0
## 19 histogram_median   0
## 20 histogram_variance 0
## 21 histogram_tendency 0
## 22 fetal_health       0
```

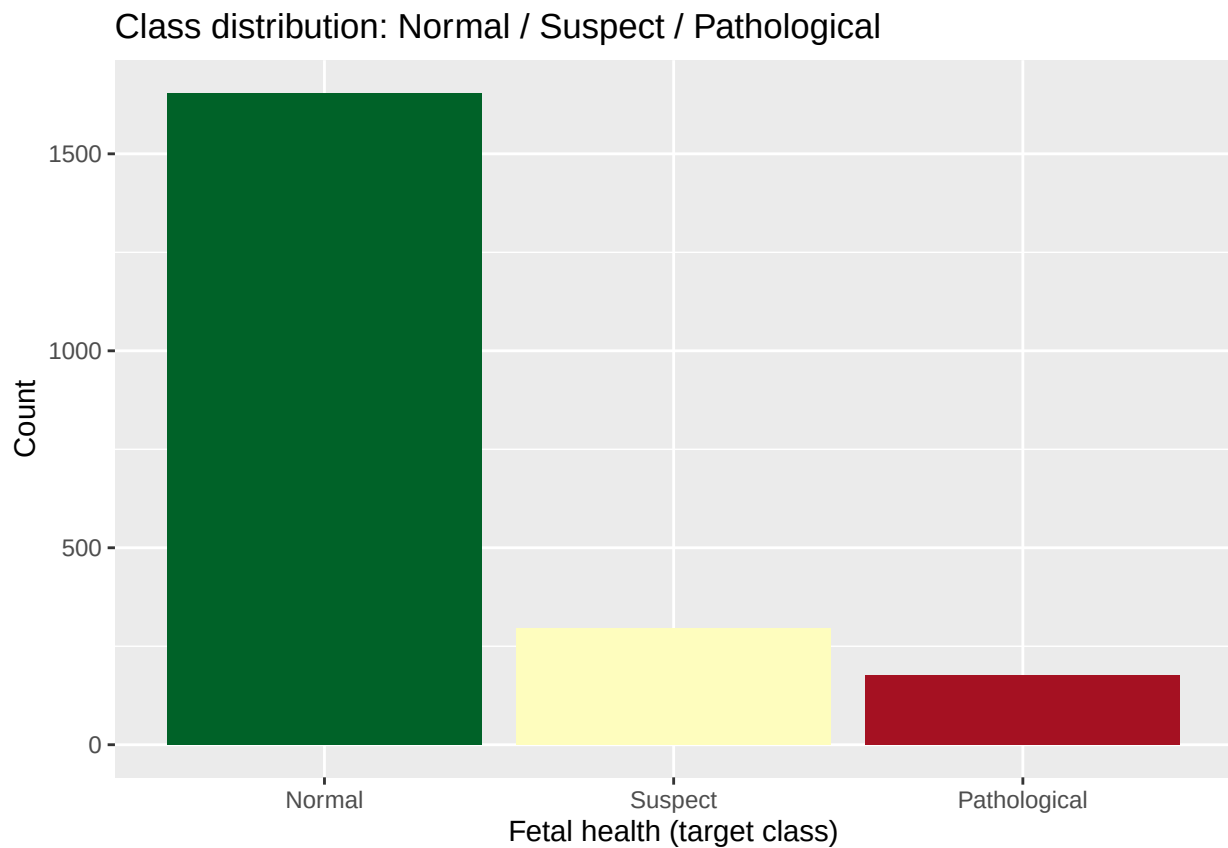
EDA

```
# outcome distribution
fetal %>%
  count(fetal_health) %>%
  mutate(prop = n / sum(n))
```

```
## # A tibble: 3 x 3
##   fetal_health     n   prop
```

```
##   <fct>         <int>  <dbl>
## 1 Normal          1655 0.778
## 2 Suspect          295 0.139
## 3 Pathological     176 0.0828
```

```
fetal %>%
  ggplot(aes(x = fetal_health, fill = fetal_health)) +
  geom_bar() +
  scale_fill_manual(
    values = rev(hcl.colors(3, "RdYlGn")),
    guide = "none"
  ) +
  labs(
    x = "Fetal health (target class)",
    y = "Count",
    title = "Class distribution: Normal / Suspect / Pathological"
  )
```



```
# summary of predictors
fetal %>%
  select(-fetal_health) %>%
  summary()
```

```
## baseline_value accelerations fetal_movement uterine_contractions
## Min. :106.0 Min. :0.000000 Min. :0.000000 Min. :0.000000
```

```

## 1st Qu.:126.0 1st Qu.:0.000000 1st Qu.:0.000000 1st Qu.:0.002000
## Median :133.0 Median :0.002000 Median :0.000000 Median :0.004000
## Mean :133.3 Mean :0.003178 Mean :0.009481 Mean :0.004366
## 3rd Qu.:140.0 3rd Qu.:0.006000 3rd Qu.:0.003000 3rd Qu.:0.007000
## Max. :160.0 Max. :0.019000 Max. :0.481000 Max. :0.015000
## light_decelerations severe_decelerations prolonged_decelerations
## Min. :0.000000 Min. :0.000e+00 Min. :0.000000
## 1st Qu.:0.000000 1st Qu.:0.000e+00 1st Qu.:0.000000
## Median :0.000000 Median :0.000e+00 Median :0.000000
## Mean :0.001889 Mean :3.293e-06 Mean :0.0001585
## 3rd Qu.:0.003000 3rd Qu.:0.000e+00 3rd Qu.:0.000000
## Max. :0.015000 Max. :1.000e-03 Max. :0.0050000
## abnormal_short_term_variability mean_value_of_short_term_variability
## Min. :12.00 Min. :0.200
## 1st Qu.:32.00 1st Qu.:0.700
## Median :49.00 Median :1.200
## Mean :46.99 Mean :1.333
## 3rd Qu.:61.00 3rd Qu.:1.700
## Max. :87.00 Max. :7.000
## percentage_of_time_with_abnormal_long_term_variability
## Min. : 0.000
## 1st Qu.: 0.000
## Median : 0.000
## Mean : 9.847
## 3rd Qu.:11.000
## Max. :91.000
## mean_value_of_long_term_variability histogram_width histogram_min
## Min. : 0.000 Min. : 3.00 Min. : 50.00
## 1st Qu.: 4.600 1st Qu.: 37.00 1st Qu.: 67.00
## Median : 7.400 Median : 67.50 Median : 93.00
## Mean : 8.188 Mean : 70.45 Mean : 93.58
## 3rd Qu.:10.800 3rd Qu.:100.00 3rd Qu.:120.00
## Max. :50.700 Max. :180.00 Max. :159.00
## histogram_max histogram_number_of_peaks histogram_number_of_zeroes
## Min. :122 Min. : 0.000 Min. : 0.0000
## 1st Qu.:152 1st Qu.: 2.000 1st Qu.: 0.0000
## Median :162 Median : 3.000 Median : 0.0000
## Mean :164 Mean : 4.068 Mean : 0.3236
## 3rd Qu.:174 3rd Qu.: 6.000 3rd Qu.: 0.0000
## Max. :238 Max. :18.000 Max. :10.0000
## histogram_mode histogram_mean histogram_median histogram_variance
## Min. : 60.0 Min. : 73.0 Min. : 77.0 Min. : 0.00
## 1st Qu.:129.0 1st Qu.:125.0 1st Qu.:129.0 1st Qu.: 2.00
## Median :139.0 Median :136.0 Median :139.0 Median : 7.00
## Mean :137.5 Mean :134.6 Mean :138.1 Mean : 18.81
## 3rd Qu.:148.0 3rd Qu.:145.0 3rd Qu.:148.0 3rd Qu.: 24.00
## Max. :187.0 Max. :182.0 Max. :186.0 Max. :269.00
## histogram_tendency
## Min. : -1.0000
## 1st Qu.: 0.0000
## Median : 0.0000
## Mean : 0.3203
## 3rd Qu.: 1.0000
## Max. : 1.0000

```

```

library(tidyverse)
library(stringr)

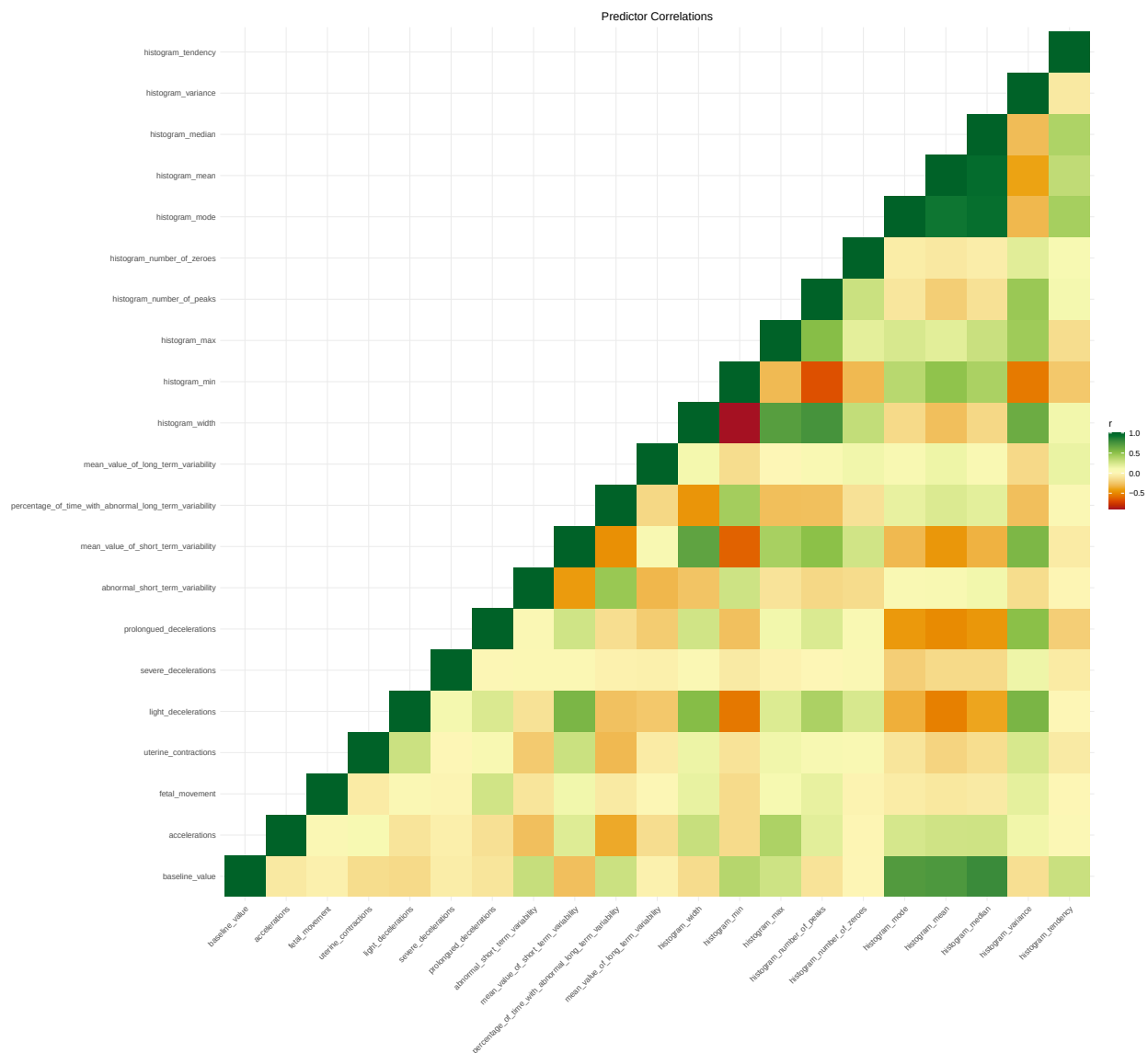
# correlation matrix (predictors only)
X <- fetal %>% select(-fetal_health)
C <- cor(X, use = "pairwise.complete.obs")

cor_long <- as.data.frame(C) %>%
  rownames_to_column("var1") %>%
  pivot_longer(-var1, names_to = "var2", values_to = "r")

cor_long <- cor_long %>%
  mutate(
    var1 = factor(var1, levels = colnames(C)),
    var2 = factor(var2, levels = colnames(C))
  ) %>%
  filter(as.integer(var1) <= as.integer(var2))

ggplot(cor_long, aes(var2, var1, fill = r)) +
  geom_tile() +
  coord_equal() +
  scale_fill_gradientn(colors = hcl.colors(20, "RdYlGn")) +
  scale_x_discrete(labels = function(x) str_wrap(x, width = 14)) +
  scale_y_discrete(labels = function(x) str_wrap(x, width = 14)) +
  labs(title = "Predictor Correlations", x = NULL, y = NULL, fill = "r") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 9),
    axis.text.y = element_text(size = 9)
  )

```



Train/test split (stratified)

```
split_obj <- initial_split(fetal, prop = 0.80, strata = fetal_health)
train_dat <- training(split_obj)
test_dat <- testing(split_obj)

train_dat %>% count(fetal_health) %>% mutate(prop = n/sum(n))
```

```
## # A tibble: 3 x 3
##   fetal_health     n  prop
##   <fct>         <int> <dbl>
## 1 Normal         1319 0.776
## 2 Suspect         238 0.14
## 3 Pathological   143 0.0841
```

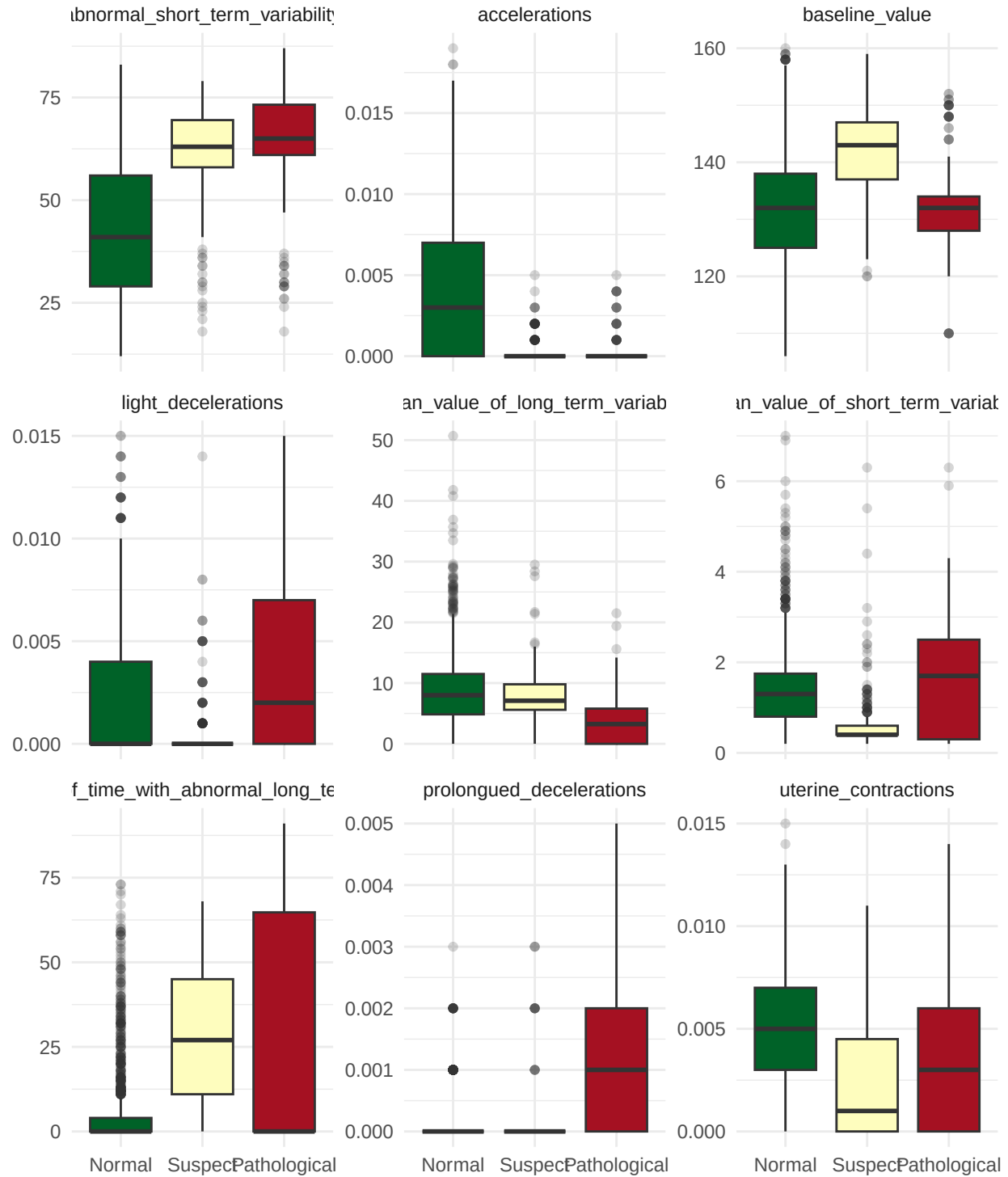
```
test_dat %>% count(fetal_health) %>% mutate(prop = n/sum(n))
```

```
## # A tibble: 3 x 3
##   fetal_health     n   prop
##   <fct>         <int> <dbl>
## 1 Normal         336 0.789
## 2 Suspect         57 0.134
## 3 Pathological    33 0.0775
```

```
vars_to_plot <- c("baseline_value",
                  "accelerations",
                  "uterine_contractions",
                  "light_decelerations",
                  "prolongued_decelerations",
                  "abnormal_short_term_variability",
                  "mean_value_of_short_term_variability",
                  "percentage_of_time_with_abnormal_long_term_variability",
                  "mean_value_of_long_term_variability")

fetal %>%
  select(fetal_health, all_of(vars_to_plot)) %>%
  pivot_longer(-fetal_health, names_to = "variable", values_to = "value") %>%
  ggplot(aes(x = fetal_health, y = value, fill = fetal_health)) +
  geom_boxplot(outlier.alpha = 0.2) +
  facet_wrap(~ variable, scales = "free_y", ncol = 3) +
  scale_fill_manual(values = rev(hcl.colors(3, "RdYlGn")), guide = "none") +
  labs(title = "Predictor distributions by class", x = NULL, y = NULL) +
  theme_minimal()
```

Predictor distributions by class



Preprocessing recipe

```
rec_scaled <- recipe(fetal_health ~ ., data = train_dat) %>%  
  step_zv(all_predictors()) %>%  
  step_normalize(all_numeric_predictors())  
  
rec_unscaled <- recipe(fetal_health ~ ., data = train_dat) %>%  
  step_zv(all_predictors())  
  
folds <- vfold_cv(train_dat, v = 5, strata = fetal_health)  
  
my_metrics <- metric_set(accuracy, bal_accuracy)
```

Fit models

Multinomial logistic regression

```
multinom_spec <- multinom_reg() %>%  
  set_mode("classification") %>%  
  set_engine("nnet", trace = FALSE)  
  
multinom_wf <- workflow() %>%  
  add_recipe(rec_scaled) %>%  
  add_model(multinom_spec)  
  
multinom_fit <- fit(multinom_wf, data = train_dat)
```

Random forest

```
rf_spec <- rand_forest(trees = 1000) %>%  
  set_mode("classification") %>%  
  set_engine("ranger", importance = "impurity")  
  
rf_wf <- workflow() %>%  
  add_recipe(rec_unscaled) %>%  
  add_model(rf_spec)  
  
rf_fit <- fit(rf_wf, data = train_dat)
```

Penalized multinomial (glmnet)

```
glmnet_spec <- multinom_reg(penalty = tune(), mixture = tune()) %>%  
  set_mode("classification") %>%  
  set_engine("glmnet")  
  
glmnet_wf <- workflow() %>%
```

```

add_recipe(rec_scaled) %>%
add_model(glmnet_spec)

set.seed(20251212)
glmnet_grid <- grid_regular(
  penalty(range = c(-4, 0)),
  mixture(range = c(0, 1)),
  levels = 6
)

glmnet_res <- tune_grid(
  glmnet_wf,
  resamples = folds,
  grid = glmnet_grid,
  metrics = my_metrics,
  control = control_grid(save_pred = TRUE)
)

```

```
## Warning: package 'Matrix' was built under R version 4.3.1
```

```
collect_metrics(glmnet_res)
```

```
## # A tibble: 72 x 8
```

	penalty	mixture	.metric	.estimator	mean	n	std_err	.config
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<int>	<dbl>	<chr>
## 1	0.0001	0	accuracy	multiclass	0.885	5	0.00492	Preprocessor1_M-
## 2	0.0001	0	bal_accuracy	macro	0.806	5	0.00944	Preprocessor1_M-
## 3	0.000631	0	accuracy	multiclass	0.885	5	0.00492	Preprocessor1_M-
## 4	0.000631	0	bal_accuracy	macro	0.806	5	0.00944	Preprocessor1_M-
## 5	0.00398	0	accuracy	multiclass	0.885	5	0.00492	Preprocessor1_M-
## 6	0.00398	0	bal_accuracy	macro	0.806	5	0.00944	Preprocessor1_M-
## 7	0.0251	0	accuracy	multiclass	0.883	5	0.00569	Preprocessor1_M-
## 8	0.0251	0	bal_accuracy	macro	0.800	5	0.0112	Preprocessor1_M-
## 9	0.158	0	accuracy	multiclass	0.859	5	0.00521	Preprocessor1_M-
## 10	0.158	0	bal_accuracy	macro	0.721	5	0.00591	Preprocessor1_M-

```
## # i 62 more rows
```

```

best_glmnet <- select_best(glmnet_res, metric = "bal_accuracy")
best_glmnet

```

```

## # A tibble: 1 x 3
##   penalty mixture .config
##   <dbl>   <dbl> <chr>
## 1 0.000631     1 Preprocessor1_Model32

```

XGboost

```

xgb_spec <- boost_tree(
  trees = tune(),
  tree_depth = tune(),

```

```

learn_rate = tune(),
loss_reduction = tune(),
sample_size = tune(),
mtry = tune(),
min_n = tune()
) %>%
  set_mode("classification") %>%
  set_engine("xgboost")

xgb_wf <- workflow() %>%
  add_recipe(rec_unscaled) %>%
  add_model(xgb_spec)

set.seed(20251212)
xgb_grid <- grid_latin_hypercube(
  trees(range = c(300, 1200)),
  tree_depth(),
  learn_rate(range = c(-4, -1)),
  loss_reduction(),
  sample_size = sample_prop(),
  mtry(range = c(3, ncol(train_dat) - 1)),
  min_n(),
  size = 25
)

```

```

## Warning: 'grid_latin_hypercube()' was deprecated in dials 1.3.0.
## i Please use 'grid_space_filling()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

```

```

xgb_res <- tune_grid(
  xgb_wf,
  resamples = folds,
  grid = xgb_grid,
  metrics = my_metrics,
  control = control_grid(save_pred = TRUE)
)

collect_metrics(xgb_res)

```

```

## # A tibble: 50 x 13
##   mtry trees min_n tree_depth learn_rate loss_reduction sample_size .metric
##   <int> <int> <int>      <int>      <dbl>      <dbl>      <dbl> <chr>
## 1    19   967     2         1  0.000113  0.00000000326    0.167 accuracy
## 2    19   967     2         1  0.000113  0.00000000326    0.167 bal_accu~
## 3     6   848    29         2  0.000272  0.00000319      0.666 accuracy
## 4     6   848    29         2  0.000272  0.00000319      0.666 bal_accu~
## 5     5   358    29         2  0.00112   0.178           0.875 accuracy
## 6     5   358    29         2  0.00112   0.178           0.875 bal_accu~
## 7     9   411    23         3  0.0442    0.0105          0.260 accuracy
## 8     9   411    23         3  0.0442    0.0105          0.260 bal_accu~
## 9    14   313    32         4  0.000330  1.25            0.349 accuracy

```

```
## 10      14    313     32          4    0.000330  1.25          0.349 bal_accur~
## # i 40 more rows
## # i 5 more variables: .estimator <chr>, mean <dbl>, n <int>, std_err <dbl>,
## #   .config <chr>
```

```
best_xgb <- select_best(xgb_res, metric = "bal_accuracy")
best_xgb
```

```
## # A tibble: 1 x 8
##   mtry trees min_n tree_depth learn_rate loss_reduction sample_size .config
##   <int> <int> <int>      <int>      <dbl>          <dbl>      <dbl> <chr>
## 1    15  1104     4         12      0.0709          0.144      0.613 Preprocess~
```

Evaluate models on test set

```
# Helper function: class-specific sensitivity from a confusion matrix
class_sensitivity <- function(cm) {
  m <- as.matrix(cm$table)
  tibble(
    class = rownames(m),
    sensitivity = diag(m) / rowSums(m)
  )
}

# Multinomial logistic regression performance
multinom_pred <- predict(multinom_fit, test_dat, type = "class") %>%
  bind_cols(test_dat %>% select(fetal_health))

multinom_acc <- accuracy(multinom_pred, truth = fetal_health, estimate = .pred_class)

multinom_cm <- conf_mat(multinom_pred, truth = fetal_health, estimate = .pred_class)

multinom_sens <- class_sensitivity(multinom_cm) %>%
  filter(class %in% c("Suspect", "Pathological"))

multinom_acc
```

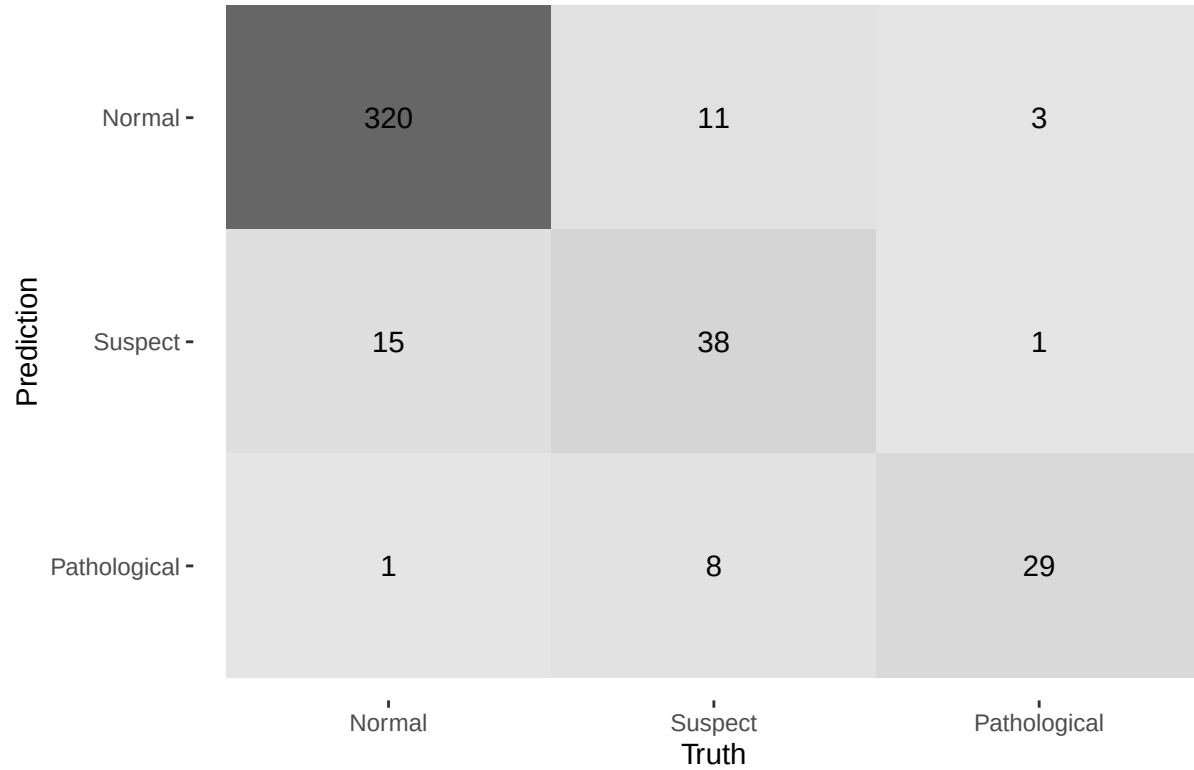
```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>    <chr>      <dbl>
## 1 accuracy multiclass    0.908
```

```
multinom_sens
```

```
## # A tibble: 2 x 2
##   class      sensitivity
##   <chr>          <dbl>
## 1 Suspect        0.704
## 2 Pathological    0.763
```

```
autoplot(multinom_cm, type = "heatmap") +
  labs(title = "Multinomial Logistic Regression: Confusion Matrix")
```

Multinomial Logistic Regression: Confusion Matrix



```
# Random forest performance
rf_pred <- predict(rf_fit, test_dat, type = "class") %>%
  bind_cols(test_dat %>% select(fetal_health))

rf_acc <- accuracy(rf_pred, truth = fetal_health, estimate = .pred_class)

rf_cm <- conf_mat(rf_pred, truth = fetal_health, estimate = .pred_class)

rf_sens <- class_sensitivity(rf_cm) %>%
  filter(class %in% c("Suspect", "Pathological"))

rf_acc
```

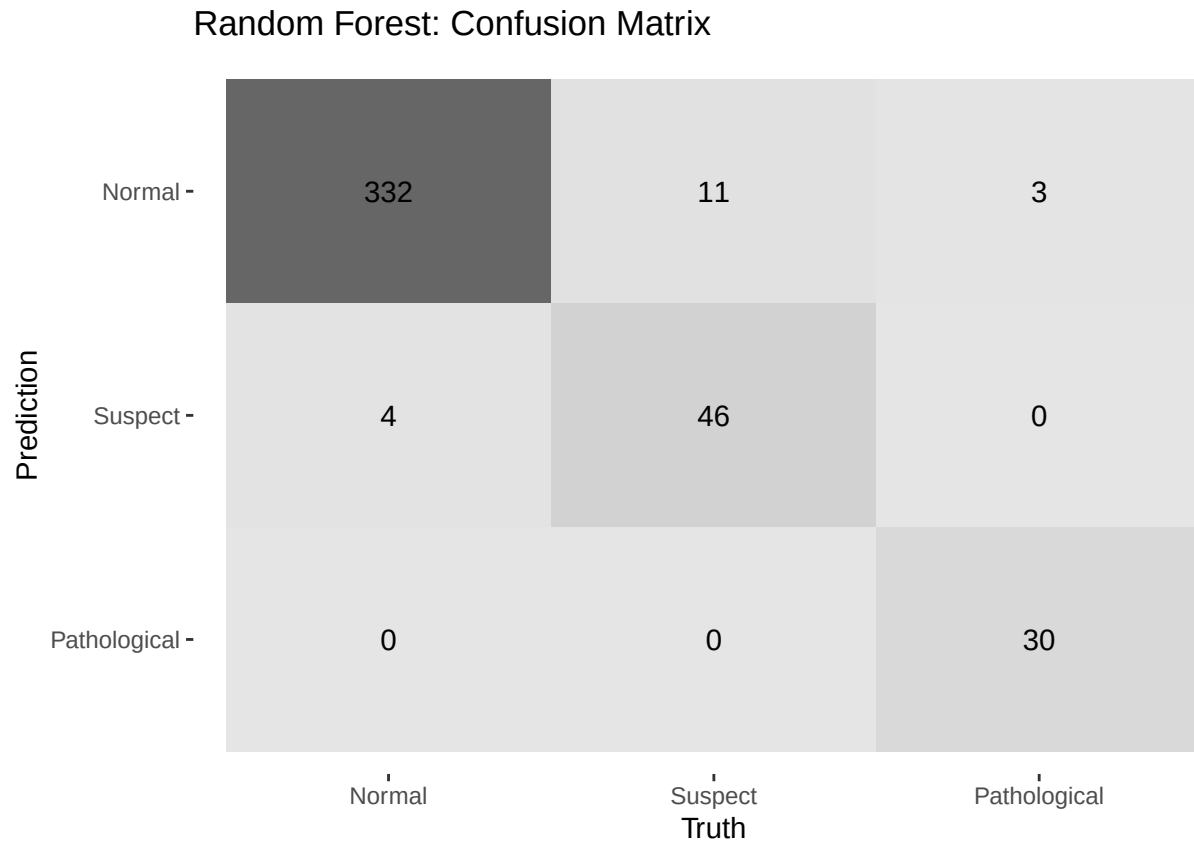
```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>      <dbl>
## 1 accuracy multiclass 0.958
```

```
rf_sens
```

```
## # A tibble: 2 x 2
```

```
##   class      sensitivity
##   <chr>      <dbl>
## 1 Suspect      0.92
## 2 Pathological    1
```

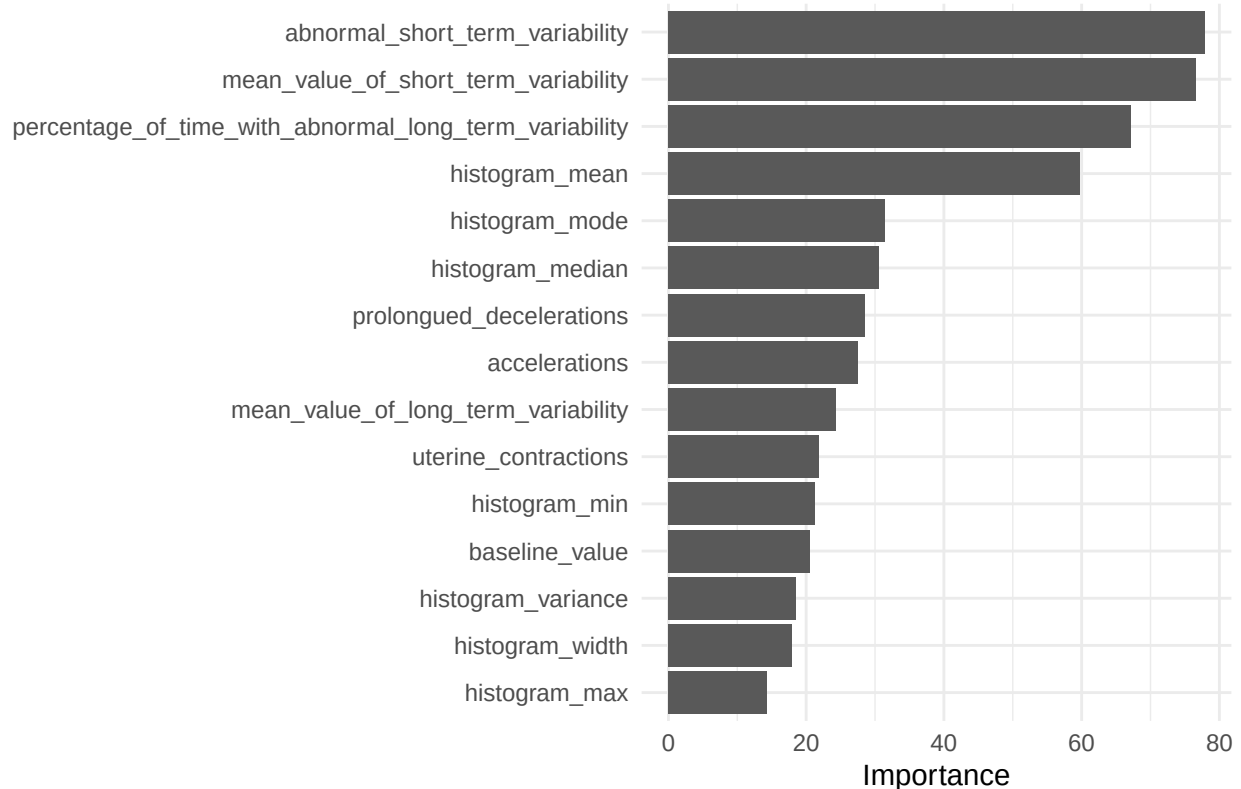
```
autoplot(rf_cm, type = "heatmap") +
  labs(title = "Random Forest: Confusion Matrix")
```



```
# Random Forest variable importance
rf_fit_obj <- extract_fit_parsnip(rf_fit)$fit
imp <- sort(rf_fit_obj$variable.importance, decreasing = TRUE)

tibble(variable = names(imp), importance = as.numeric(imp)) %>%
  slice_head(n = 15) %>%
  ggplot(aes(x = reorder(variable, importance), y = importance)) +
  geom_col() +
  coord_flip() +
  labs(title = "Random Forest: Top 15 variable importances", x = NULL, y = "Importance") +
  theme_minimal()
```

Random Forest: Top 15 variable importance



```
# glmnet performance
glmnet_final_wf <- finalize_workflow(glmnet_wf, best_glmnet)
glmnet_fit <- fit(glmnet_final_wf, data = train_dat)

glmnet_pred <- predict(glmnet_fit, test_dat, type = "class") %>%
  bind_cols(test_dat %>% select(fetal_health))

glmnet_acc <- accuracy(glmnet_pred, truth = fetal_health, estimate = .pred_class)
glmnet_cm <- conf_mat(glmnet_pred, truth = fetal_health, estimate = .pred_class)
glmnet_sens <- class_sensitivity(glmnet_cm) %>%
  filter(class %in% c("Suspect", "Pathological"))

glmnet_acc
```

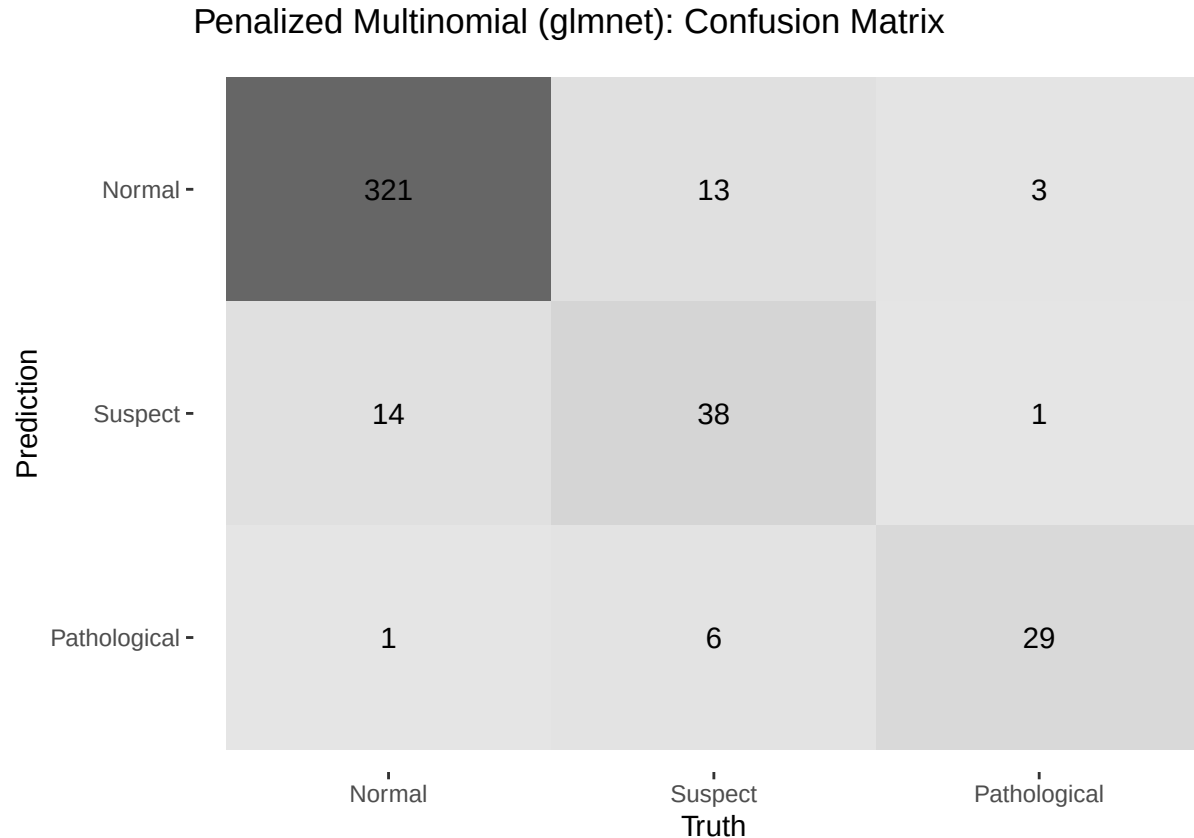
```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>      <dbl>
## 1 accuracy multiclass 0.911
```

```
glmnet_sens
```

```
## # A tibble: 2 x 2
##   class      sensitivity
##   <chr>          <dbl>
## 1 Suspect      0.717
```

```
## 2 Pathological          0.806
```

```
autoplot(glmnet_cm, type = "heatmap") +  
  labs(title = "Penalized Multinomial (glmnet): Confusion Matrix")
```



```
# XGBoost performance  
xgb_final_wf <- finalize_workflow(xgb_wf, best_xgb)  
xgb_fit <- fit(xgb_final_wf, data = train_dat)  
  
xgb_pred <- predict(xgb_fit, test_dat, type = "class") %>%  
  bind_cols(test_dat %>% select(fetal_health))  
  
xgb_acc <- accuracy(xgb_pred, truth = fetal_health, estimate = .pred_class)  
xgb_cm <- conf_mat(xgb_pred, truth = fetal_health, estimate = .pred_class)  
xgb_sens <- class_sensitivity(xgb_cm) %>%  
  filter(class %in% c("Suspect", "Pathological"))  
  
xgb_acc
```

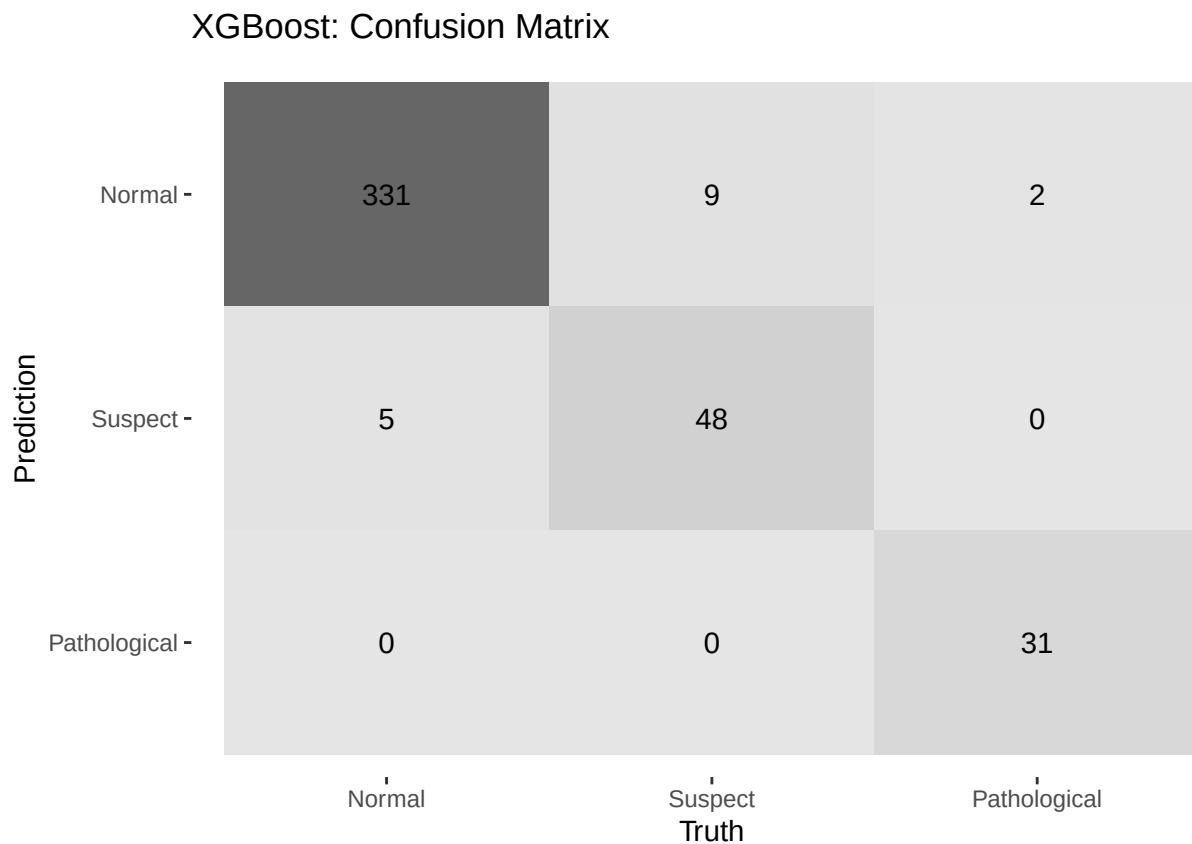
```
## # A tibble: 1 x 3  
##   .metric .estimator .estimate  
##   <chr>   <chr>      <dbl>  
## 1 accuracy multiclass 0.962
```



```
xgb_sens
```

```
## # A tibble: 2 x 2
##   class      sensitivity
##   <chr>         <dbl>
## 1 Suspect      0.906
## 2 Pathological    1
```

```
autoplot(xgb_cm, type = "heatmap") +
  labs(title = "XGBoost: Confusion Matrix")
```



Side-by-side model comparison

```
bind_rows(
  multinom_acc %>% mutate(model = "Multinomial (nnet)"),
  rf_acc %>% mutate(model = "Random forest"),
  glmnet_acc %>% mutate(model = "Penalized multinomial (glmnet)"),
  xgb_acc %>% mutate(model = "XGBoost")
) %>%
  select(model, .metric, .estimate) %>%
  arrange(desc(.estimate))
```

```
## # A tibble: 4 x 3
##   model                .metric .estimate
##   <chr>                <chr>      <dbl>
## 1 XGBoost              accuracy    0.962
## 2 Random forest        accuracy    0.958
## 3 Penalized multinomial (glmnet) accuracy    0.911
## 4 Multinomial (nnet)    accuracy    0.908
```

```
bind_rows(
  multinom_sens %>% mutate(model = "Multinomial (nnet)"),
  rf_sens       %>% mutate(model = "Random forest"),
  glmnet_sens   %>% mutate(model = "Penalized multinomial (glmnet)"),
  xgb_sens      %>% mutate(model = "XGBoost")
) %>%
  select(model, class, sensitivity) %>%
  arrange(class, desc(sensitivity))
```

```
## # A tibble: 8 x 3
##   model                class      sensitivity
##   <chr>                <chr>      <dbl>
## 1 Random forest        Pathological    1
## 2 XGBoost              Pathological    1
## 3 Penalized multinomial (glmnet) Pathological    0.806
## 4 Multinomial (nnet)    Pathological    0.763
## 5 Random forest        Suspect        0.92
## 6 XGBoost              Suspect        0.906
## 7 Penalized multinomial (glmnet) Suspect        0.717
## 8 Multinomial (nnet)    Suspect        0.704
```